

## ASSIGNMENT 5: MEASURE THEORY

1. Give an example where

$$D^+(f+g) \neq D^+f + D^+g.$$

2. Let

$$f(x) = \begin{cases} ax \sin^2\left(\frac{1}{x}\right) + bx \cos^2\left(\frac{1}{x}\right), & x > 0, \\ 0, & x = 0, \\ a'x \sin^2\left(\frac{1}{x}\right) + b'x \cos^2\left(\frac{1}{x}\right), & x < 0, \end{cases}$$

where  $a < b$  and  $a' < b'$ . Assume that  $f$  is continuous at  $x = 0$ .

Evaluate the four Dini derivatives of  $f$  at  $x = 0$ , namely

$$D^+f(0), \quad D_+f(0), \quad D^-f(0), \quad D_-f(0).$$

3. Define  $f : [0, 1] \rightarrow \mathbb{R}$  by

$$f(0) = 0, \quad f(x) = x^2 \sin\left(\frac{1}{x}\right), \quad x > 0,$$

Show that  $f \in BV([0, 1])$ , i.e.,  $f$  is of bounded variation on  $[0, 1]$ .

4. Suppose  $f : [a, b] \rightarrow \mathbb{R}$  is of bounded variation and continuous. Prove that  $f = f_1 - f_2$ , where both  $f_1$  and  $f_2$  are monotonic and continuous.
5. Let  $f \in L^1([a, b])$ , and define

$$F(x) := \int_{[a, x]} f \, d\lambda, \quad x \in [a, b].$$

Show that

$$V_a^b(F) = \int_{[a, b]} |f| \, d\lambda.$$

6. Show that if  $f \in BV[a, b]$ , then:

(a)  $\int_{[a, b]} |f'| \, d\lambda \leq V_a^b(f).$

(b)  $\int_{[a, b]} |f'| \, d\lambda = V_a^b(f)$  if, and only if  $f$  is absolutely continuous.

7. Let  $f : [a, b] \rightarrow \mathbb{R}$  be an absolutely continuous function. Define

$$h(x) := V_a^x(f), \quad x \in [a, b].$$

Show that  $h$  is absolutely continuous. (Hint: use 2nd fundamental theorem)

8. Let  $g : [a, b] \rightarrow [c, d]$  be absolutely continuous and monotone, and let  $f : [c, d] \rightarrow \mathbb{R}$  be absolutely continuous. Show that  $f \circ g$  is absolutely continuous on  $[a, b]$ .

9. Define two functions  $f$  and  $g$  on  $[0, 1]$  as follows:

$$f(x) := \begin{cases} x^2 \sin^2\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

and let

$$g(x) := \sqrt{x}.$$

Show that:

- (i)  $f$  is absolutely continuous on  $[0, 1]$ .
- (ii)  $g$  is absolutely continuous, but not Lipschitz continuous, on  $[0, 1]$ .
- (iii)  $g \circ f$  is not absolutely continuous on  $[0, 1]$ .

10. Show that absolutely continuous functions map null sets to null sets.

11. Let  $f \in L^1(\mathbb{R})$  be real-valued. Show that for a.e.  $x \in \mathbb{R}$ ,

$$f(x) = \lim_{h \rightarrow 0^+} \frac{1}{2h} \int_{[x-h, x+h]} f \, d\lambda.$$

(Hint: First prove the result for simple functions. Then extend it to general  $f \in L^1(\mathbb{R})$  using density of simple functions in  $L^1$ . In the process, you may need Tonelli's theorem, which allows you to interchange the order of integration in a double integral when the integrand is non-negative.)

12. Let  $A \subset [0, 1]$  be a Borel set such that

$$0 < \lambda(A \cap I) < \lambda(I)$$

for every subinterval  $I \subset [0, 1]$ . Define

$$F(x) = \lambda([0, x] \cap A), \quad x \in [0, 1].$$

Show that  $F$  is absolutely continuous and strictly increasing, but  $F' = 0$  on a set of positive Lebesgue measure.

13. Assume that  $f$  is absolutely continuous on  $[a, b]$  and  $f' \in L^p([a, b])$  for some  $1 < p < \infty$ . Prove that  $f$  is Hölder continuous with exponent  $\frac{1}{p'}$ . (Hint: use 2nd fundamental theorem and Hölder's inequality)